Average Rate of Change

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Computing average rate of change**

Find the average rate of change of $f(x) = x^2$ over each interval:

a $[0, 2]$ $= 2$ **b** $[1, 3]$ $= 4$ **c** $[2, 5]$ $= 7$ **d** $[-1, 1]$ $= 0$

(a) $= 2$; (b) $= 4$; (c) $= 7$; (d) $= 0$

2 **[Foundational]** • **Target: Computing average rate of change**

Find the average rate of change of $f(x) = 2^x$ over the interval $[0, 3]$.

$\frac{f(3) - f(0)}{3 - 0} = \frac{8 - 1}{3} = \frac{7}{3}$.

3 **[Foundational]** • **Target: Computing average rate of change**

Show that the average rate of change of $g(x) = 3x - 5$ is the same over any interval $[a, b]$, and state its value.

$\frac{g(b) - g(a)}{b - a} = \frac{(3b - 5) - (3a - 5)}{b - a} = \frac{3(b - a)}{b - a} = 3$. A linear function changes at a constant rate equal to its slope.

4 **[Foundational]** • **Target: Applications**

A ball is thrown upward so that its height after t seconds is $h(t) = -5t^2 + 20t$ meters.

a Find the average rate of change of the height over $[0, 2]$. $\frac{h(2) - h(0)}{2 - 0} = \frac{20 - 0}{2} = 10$ m/s.

b Find the average rate of change of the height over $[1, 3]$. $h(1) = 15$ and $h(3) = 15$, so the average rate of change is $\frac{15 - 15}{2} = 0$ m/s.

c Interpret your answer to part b in the context of the ball's motion. Over $[1, 3]$ the ball rises and then falls back to the same height, so on average its height does not change.

(a) $\frac{h(2) - h(0)}{2 - 0} = \frac{20 - 0}{2} = 10$ m/s.; (b) $h(1) = 15$ and $h(3) = 15$, so the average rate of change is $\frac{15 - 15}{2} = 0$ m/s.; (c) Over $[1, 3]$ the ball rises and then falls back to the same height, so on average its height does not change.

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5 [Foundational] • Target: Applications

The graph of $f(x) = -\frac{1}{2}x^2 + 2x + 1$ is shown. Estimate the average rate of change over $[0, 4]$, then verify your estimate algebraically.

$f(0) = 1$ and $f(4) = -8 + 8 + 1 = 1$, so the average rate of change over $[0, 4]$ is $\frac{1-1}{4} = 0$. The secant line through the endpoints is horizontal.

6 [Foundational] • Target: Applications

A company's revenue (in thousands of riyals) is modeled by $R(t) = 40 + 6t - 0.5t^2$, where t is years since 2020.

a Find the average rate of change of revenue over $[0, 4]$. $R(0) = 40$, $R(4) = 40 + 24 - 8 = 56$: rate $= \frac{56-40}{4} = 4$ thousand riyals/year.

b Find the average rate of change of revenue over $[4, 8]$. $R(8) = 40 + 48 - 32 = 56$: rate $= \frac{56-56}{4} = 0$ thousand riyals/year.

c Compare your two answers and interpret what they say about the company's growth. Revenue grew steadily during the first period but plateaued during the second — the average growth rate dropped to zero.

(a) $R(0) = 40$, $R(4) = 40 + 24 - 8 = 56$: rate $= \frac{56-40}{4} = 4$ thousand riyals/year.; (b) $R(8) = 40 + 48 - 32 = 56$: rate $= \frac{56-56}{4} = 0$ thousand riyals/year.; (c) Revenue grew steadily during the first period but plateaued during the second — the average growth rate dropped to zero.

7 [Foundational] • Target: Comparing rates and estimating from a table

A table of values for $g(x)$ is given: $g(1) = 5$, $g(3) = 17$, $g(6) = 44$, $g(10) = 92$. Find the average rate of change of g over:

a $[1, 3]$ $= \frac{17-5}{2} = 6$ **b** $[3, 6]$ $= \frac{44-17}{3} = 9$ **c** $[6, 10]$ $= \frac{92-44}{4} = 12$

(a) $= \frac{17-5}{2} = 6$; (b) $= \frac{44-17}{3} = 9$; (c) $= \frac{92-44}{4} = 12$

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: Comparing rates and estimating from a table

Based on the previous question, is the average rate of change of g increasing, decreasing, or constant as x increases? What does this suggest about the shape of g's graph?

The average rate of change is increasing $(6, 9, 12)$, suggesting g is concave up — consistent with g being a quadratic or otherwise increasingly steep function.

9 [Intermediate] • Target: Comparing rates and estimating from a table

Two functions model distance traveled: $d_1(t) = 5t$ and $d_2(t) = t^2$ (in km, t in hours). Find the average speed (average rate of change) of each over $[2, 6]$, and state which traveler was faster on average.

d_1: $\frac{30-10}{4} = 5$ km/h. d_2: $\frac{36-4}{4} = 8$ km/h. The second traveler had the greater average speed over this interval.

10 [Intermediate] • Target: Average rate of change and secant lines

Explain the geometric meaning of the average rate of change of f over $[a, b]$ in terms of the secant line through $(a, f(a))$ and $(b, f(b))$.

The average rate of change equals the slope of the secant line connecting the two points $(a, f(a))$ and $(b, f(b))$ on the graph of f.

11 [Intermediate] • Target: Average rate of change and secant lines

For $f(x) = x^3$, find the average rate of change over $[-1, 2]$, and write the equation of the secant line through the corresponding points.

$f(-1) = -1$, $f(2) = 8$. Rate $= \frac{8-(-1)}{2-(-1)} = \frac{9}{3} = 3$. Secant line: $y - 8 = 3(x - 2)$, i.e. $y = 3x + 2$.

12 [Intermediate] • Target: Comparing average rate of change across different functions

Find the average rate of change of $f(x) = 2^x$ and $g(x) = x^2$ over $[0, 4]$, and determine which function grew faster on average.

f: $\frac{16-1}{4} = 3.75$. g: $\frac{16-0}{4} = 4$. Over this interval, $g(x) = x^2$ had the slightly larger average rate of change.

13 [Intermediate] • Target: Comparing average rate of change across different functions

Find the average rate of change of $f(x) = 2^x$ over $[4, 8]$ and compare it to its value over $[0, 4]$ (found above). What does this reveal about exponential growth?

$f(4) = 16$, $f(8) = 256$. Rate $= \frac{256-16}{4} = 60$. This is much larger than the rate over $[0, 4]$ (3.75), illustrating that exponential functions grow at an accelerating average rate over successive equal-width intervals, unlike linear functions.

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14 **[Intermediate]** • **Target: Average rate of change from a graph**

Using the graph of $f(x) = x^2 - 4x + 3$ shown, estimate the average rate of change over $[0, 4]$ by reading endpoint values from the graph, then confirm algebraically.

From the graph: $f(0) = 3$, $f(4) = 3$. Average rate of change $= \frac{3-3}{4-0} = 0$, matching the fact that the parabola's vertex lies at the midpoint $x = 2$ and the two endpoints have equal height.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Units and interpretation in context** • **Cross-Lesson: 6.01 Polar Coordinates and Conversions**

A tank's water volume is $V(t) = 200 - 3t^2$ liters, where t is minutes. Find the average rate of change over $[0, 5]$, including correct units, and interpret its sign.

$\frac{V(5) - V(0)}{5 - 0} = \frac{125 - 200}{5} = -15$ liters/minute. The negative sign means the tank is draining on average at 15 liters per minute over this interval.

16 **[Challenging]** • **Target: Units and interpretation in context** • **Cross-Lesson: 6.02 Polar Equations and Graphs**

A population model gives $P(t) = 500e^{0.1t}$ (people, years). Find the average rate of change over $[0, 10]$ and state its units, rounding to the nearest whole number.

$P(0) = 500$, $P(10) = 500e^1 \approx 1359$. Average rate of change $\approx \frac{1359 - 500}{10} \approx 86$ people/year.

17 **[Challenging]** • **Target: Comparing rates and estimating from a table** • **Cross-Lesson: 7.01 Exponential Functions and Their Graphs**

Given vectors $\mathbf{a} = 2\mathbf{i} - 5\mathbf{j}$ and $\mathbf{b} = -3\mathbf{i} + 4\mathbf{j}$, find the scalar projection of $\mathbf{a}$ onto $\mathbf{b}$ and the angle θ between them to the nearest tenth of a degree.

Calculate the dot product: $\mathbf{a} \cdot \mathbf{b} = (2)(-3) + (-5)(4) = -6 - 20 = -26.$ Calculate the magnitudes: $|\mathbf{a}| = \sqrt{2^2 + (-5)^2} = \sqrt{4 + 25} = \sqrt{29} \approx 5.385,$ $|\mathbf{b}| = \sqrt{(-3)^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5.$ Scalar projection of $\mathbf{a}$ onto $\mathbf{b}$: $\text{comp}_{\mathbf{b}}\mathbf{a} = \frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{b}|} = \frac{-26}{5} = -5.2.$ Angle θ between vectors.

18 **[Challenging]** • **Target: Average rate of change and secant lines** • **Cross-Lesson: 7.02 Logarithms and Their Properties**

Eliminate the parameter to determine the Cartesian equation of the curve $x = 1 + 2\sec t$, $y = -2 + 3\tan t$. State the shape, center, and vertices.

Isolate the trigonometric terms: $\sec t = \frac{x-1}{2}, \quad \tan t = \frac{y+2}{3}.$ Apply the fundamental Pythagorean identity $\sec^2 t - \tan^2 t = 1$: $\left(\frac{x-1}{2}\right)^2 - \left(\frac{y+2}{3}\right)^2 = 1 \Rightarrow \frac{(x-1)^2}{4} - \frac{(y+2)^2}{9} = 1.$ This is a horizontal hyperbola with:Center: $(h, k) = (1, -2)$Transverse semi-axis: $a = 2$Vertices: $(h \pm a, k) = (1 \pm 2, -2) = (-1, -2)$ and $(3, -2)$

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19 [Challenging] • Target: Comparing average rate of change across different functions • Cross-Lesson: 7.03 Exponential and Logarithmic Equations

Find the average rate of change of the polynomial function $f(x) = 2x^3 - 5x + 4$ over the closed interval $[-1, 2]$.

Step 1: Evaluate the function at the interval endpoints: $f(-1) = 2(-1)^3 - 5(-1) + 4 = -2 + 5 + 4 = 7$. $f(2) = 2(2)^3 - 5(2) + 4 = 16 - 10 + 4 = 10$. Step 2: Compute the difference quotient: AROC $= \frac{f(2) - f(-1)}{2 - (-1)} = \frac{10 - 7}{3} = \frac{3}{3} = \mathbf{1}$.

20 [Challenging] • Target: Average rate of change from a graph • Cross-Lesson: 7.04 Applications: Growth and Decay

[Vectors, Lines & Parametric Kinematics · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Computing average rate of change

In triangle ABC, angle A is 52 degrees. The side opposite angle B (AC) has a length of 10 units, and the side opposite angle C (AB) has a length of 14 units. What is the length of the side opposite angle A (BC)?

a. 10.3

b. **12.1** ✓

c. 14.8

d. 8.5

We can use the Law of Cosines to solve for the unknown side BC. The Law of Cosines states that for any triangle with sides a, b, and c, and opposite angles A, B, and C respectively: $a^2 = b^2 + c^2 - 2bc\cos(A)$. In this problem, we have: side b = AC = 10, side c = AB = 14, and angle A = 52 degrees. We want to find side a = BC. Substituting the values into the formula: $a^2 = 10^2 + 14^2 - 2(10)(14)\cos(52^\circ)$ $a^2 = 100 + 196 - 280\cos(52^\circ)$ $a^2 = 296 - 280(0.61566)$ $a^2 = 296 - 172.3848$.

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22 [Foundational] • Target: Applications

A circle has a radius of 6 units. A chord within this circle subtends a sector with a central angle of 60 degrees. What is the length of the chord?

a. 6 units ✓

b. $6\sqrt{3}$ units

c. 12 units

d. $12\sqrt{3}$ units

We have a circle with radius $r = 6$ and a sector with a central angle of 60 degrees. The chord connecting the endpoints of the arc of this sector forms an isosceles triangle with the two radii. Since the central angle is 60 degrees and the other two angles in the triangle must be equal (because the sides opposite them are radii), each of the other angles is $(180^\circ - 60^\circ)/2 = 60^\circ$. Therefore, the triangle is equilateral. In an equilateral triangle, all sides are equal. Since two sides are radii of.

23 [Foundational] • Target: Comparing rates and estimating from a table

In a right-angled triangle ABC, the angle B is 90 degrees. If the length of side AB is 8 units and the length of side BC is 6 units, what is the length of the hypotenuse AC?

a. 10 units ✓

b. 14 units

c. 48 units

d. 20 units

This is a right-angled triangle, so we can use the Pythagorean theorem: $a^2 + b^2 = c^2$. Here, the sides are $AB = 8$ and $BC = 6$, and the hypotenuse is AC. So, $AC^2 = AB^2 + BC^2$. $AC^2 = 8^2 + 6^2$ $AC^2 = 64 + 36$ $AC^2 = 100$ $AC = \sqrt{100}$ $AC = 10$ units. Common mistakes include adding the sides (8 + 6 = 14) or multiplying them (8 * 6 = 48) instead of applying the Pythagorean theorem.

24 [Foundational] • Target: Average rate of change and secant lines

$f(x) = 2x^2 + 2x - 5$. Find the average rate of change of f on $[1, 3]$.

a. 10 ✓

b. -10

c. 9

d. 5

$\text{AROC} = \frac{f(3) - f(1)}{3 - (1)} = \frac{19 - (-1)}{2} = 10.$

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25 [Foundational] • Target: Comparing average rate of change across different functions

Two particles, A and B, move in the xy-plane. Their position vectors at time t seconds, for $t \geq 0$, are given by $\mathbf{r}_A(t) = (2t; 4t - 1)$ and $\mathbf{r}_B(t) = (3 + 5t; 1 - 2t)$ respectively. The particles start their motion at $t = 0$. Find the minimum distance between particle A and particle B.

a. $\sqrt{26}$ ✓

b. $2\sqrt{13}$

c. $3\sqrt{10}$

d. $4\sqrt{5}$

The position vectors of particle A and particle B at time t are given by $\mathbf{r}_A(t) = (2t; 4t - 1)$ and $\mathbf{r}_B(t) = (3 + 5t; 1 - 2t)$. The vector representing the displacement from particle B to particle A at time t is $\mathbf{d}(t) = \mathbf{r}_A(t) - \mathbf{r}_B(t) = (2t - (3 + 5t); (4t - 1) - (1 - 2t)) = (-3t - 3; 6t - 2)$. The distance between the particles at time t is the magnitude of this displacement vector.

26 [Foundational] • Target: Average rate of change from a graph

Consider a circle with its center at the origin $(0, 0)$ and a radius of 5 units. A chord of this circle is perpendicular to the x-axis and intersects the x-axis at the point $(3, 0)$. The endpoints of this chord on the circle are A and B. Determine the length of the chord AB.

a. 6

b. **8** ✓

c. 10

d. 4

The equation of the circle centered at the origin with radius 5 is $x^2 + y^2 = 5^2 = 25$. The chord is perpendicular to the x-axis and intersects it at $(3, 0)$. This means the x-coordinate of the endpoints of the chord, A and B, is 3. Substitute $x = 3$ into the circle's equation to find the corresponding y-coordinates: $3^2 + y^2 = 25$ $9 + y^2 = 25$ $y^2 = 25 - 9$ $y^2 = 16$ $y = \pm 4$ So the endpoints of the chord are $A = (3, 4)$ and $B = (3, -4)$. The length of the chord AB is the distance between these t.

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27 [Foundational] • Target: Units and interpretation in context

The graph of the function $f(x) = a \cdot b^x$ passes through the points $(1, 12)$ and $(3, 108)$. Determine the value of the parameter a.

a. 1

b. **2 ✓**

c. 3

d. 4

We are given the function $f(x) = a \cdot b^x$ and two points it passes through: $(1, 12)$ and $(3, 108)$. Using the first point $(1, 12)$, we have: $12 = a \cdot b^1$ (Equation 1) Using the second point $(3, 108)$, we have: $108 = a \cdot b^3$ (Equation 2) To find a and b, we can divide Equation 2 by Equation 1: $\frac{108}{12} = \frac{a \cdot b^3}{a \cdot b^1}$ $9 = b^2$ Taking the square root of both sides, and assuming $b > 0$ for exponential functions: $b = 3$ Now, substitute the value of $b = 3$ back into E.

28 [Foundational] • Target: Computing average rate of change

Consider the exponential growth model $P(t) = P_0 e^{kt}$, where $P(t)$ is the population at time t and P_0 is the initial population. If the initial population is $P_0 = 500$ and after 10 years the population is $P(10) = 1200$, find the annual growth rate k. Round your answer to four decimal places.

a. 0.0588

b. 0.0875

c. 0.1200

d. **0.0434 ✓**

We are given the exponential growth model $P(t) = P_0 e^{kt}$. We know that the initial population $P_0 = 500$ and that after 10 years, $P(10) = 1200$. We can substitute these values into the model: $1200 = 500e^{k \cdot 10}$ Divide both sides by 500: $\frac{1200}{500} = e^{10k}$ $2.4 = e^{10k}$ To solve for k, we take the natural logarithm of both sides: $\ln(2.4) = \ln(e^{10k})$ $\ln(2.4) = 10k$ Now, divide by 10: $k = \frac{\ln(2.4)}{10}$ Using a calculator, $\ln(2.4) \approx 0.8754687$. So.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Applications

Consider two vectors, $\mathbf{u} = (3; k)$ and $\mathbf{v} = (-2; 4)$. If $\mathbf{u}$ is parallel to $\mathbf{v}$, find the value of the scalar parameter k.

a. **-6** ✓

b. 6

c. 3

d. -3

Two vectors are parallel if one is a scalar multiple of the other. Therefore, there must exist a scalar c such that $\mathbf{u} = c\mathbf{v}$. This gives us the equations: $3 = c(-2)$ $k = c(4)$ From the first equation, we can solve for c: $c = \frac{3}{-2} = -\frac{3}{2}$ Now substitute this value of c into the second equation to find k: $k = \left(-\frac{3}{2}\right)(4)$ $k = -\frac{12}{2}$ $k = -6$ Therefore, the value of k is -6. A common algebraic trap is to mistakenly set the components equal with.

30 [Intermediate] • Target: Comparing rates and estimating from a table

In triangle ABC, side $a = 10$ units and side $b = 15$ units. If the angle $C = 60°$, find the length of side c. Round your answer to two decimal places.

a. **13.23** ✓

b. 14.70

c. 12.96

d. 15.52

We can use the Law of Cosines to find the length of side c. The Law of Cosines states that $c^2 = a^2 + b^2 - 2ab\cos C$. Given: $a = 10$, $b = 15$, and $C = 60°$. Substitute the values into the formula: $c^2 = 10^2 + 15^2 - 2(10)(15)\cos 60°$ $c^2 = 100 + 225 - 300 \cdot \frac{1}{2}$ $c^2 = 325 - 150$ $c^2 = 175$ $c = \sqrt{175}$ $c = \sqrt{25 \cdot 7}$ $c = 5\sqrt{7}$ To find the decimal approximation, we calculate: $c \approx 5 \cdot 2.64575$ $c \approx 13.22875$ Rounding to two decimal places.

31 [Intermediate] • Target: Average rate of change and secant lines

Consider two points in the Cartesian plane, $P = (2, 3)$ and $Q = (5, 7)$. What is the component form of the vector $\vec{PQ}$?

a. $\langle 3, 4 \rangle$ ✓

b. $\langle -3, -4 \rangle$

c. $\langle 7, 10 \rangle$

d. $\langle 1, 4 \rangle$

To find the component form of a vector $\vec{PQ}$ from point $P(x_1, y_1)$ to point $Q(x_2, y_2)$, we subtract the coordinates of the initial point P from the coordinates of the terminal point Q. The formula is $\vec{PQ} = \langle x_2 - x_1, y_2 - y_1 \rangle$. Given $P = (2, 3)$ and $Q = (5, 7)$, we have $x_1 = 2$, $y_1 = 3$, $x_2 = 5$, and $y_2 = 7$. Applying the formula: $\vec{PQ} = \langle 5 - 2, 7 - 3 \rangle$ $\vec{PQ} = \langle 3, 4 \rangle$ Therefore, the component form of the vector $\vec{PQ}$ is.

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32 [Intermediate] • Target: Comparing average rate of change across different functions

Two vectors are given by $\vec{u} = 3\hat{i} + k\hat{j}$ and $\vec{v} = -\hat{i} + 4\hat{j}$. If $\vec{u}$ and $\vec{v}$ are orthogonal, find the value of the scalar parameter k.

a. 3/4

b. **4/3 ✓**

c. -3/4

d. -4/3

Two vectors $\vec{u}$ and $\vec{v}$ are orthogonal if and only if their dot product is zero. The dot product of $\vec{u} = u_1\hat{i} + u_2\hat{j}$ and $\vec{v} = v_1\hat{i} + v_2\hat{j}$ is given by $\vec{u} \cdot \vec{v} = u_1v_1 + u_2v_2$. Given $\vec{u} = 3\hat{i} + k\hat{j}$ and $\vec{v} = -\hat{i} + 4\hat{j}$, their dot product is: $\vec{u} \cdot \vec{v} = (3)(-1) + (k)(4)$ $\vec{u} \cdot \vec{v} = -3 + 4k$ For the vectors to be orthogonal, their dot product must be zero: $-3 + 4k = 0$ $4k = 3$.

33 [Intermediate] • Target: Average rate of change from a graph

Consider the vector $\mathbf{v} = \langle 3, 4 \rangle$. What is the magnitude of this vector?

a. **5 ✓**

b. 7

c. 12

d. 25

The magnitude of a vector $\mathbf{v} = \langle x, y \rangle$ is given by the formula $|\mathbf{v}| = \sqrt{x^2 + y^2}$. For the given vector $\mathbf{v} = \langle 3, 4 \rangle$, we have x = 3 and y = 4. Substituting these values into the formula: $|\mathbf{v}| = \sqrt{3^2 + 4^2}$ $|\mathbf{v}| = \sqrt{9 + 16}$ $|\mathbf{v}| = \sqrt{25}$ $|\mathbf{v}| = 5$ Therefore, the magnitude of the vector $\mathbf{v}$ is 5.

34 [Intermediate] • Target: Units and interpretation in context

Consider the vector $\mathbf{v}$ with initial point $P = (2, 3)$ and terminal point $Q = (5, 7)$. What are the components of the vector $\mathbf{v}$?

a. **■3, 4■ ✓**

b. ■7, 10■

c. ■-3, -4■

d. ■4, 3■

The components of a vector from point $P = (x_1, y_1)$ to point $Q = (x_2, y_2)$ are given by $\langle x_2 - x_1, y_2 - y_1 \rangle$. In this case, $P = (2, 3)$ and $Q = (5, 7)$. Therefore, the components of vector $\mathbf{v}$ are $\langle 5 - 2, 7 - 3 \rangle = \langle 3, 4 \rangle$.

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35 [Intermediate] • Target: Computing average rate of change

Find the period and phase shift of $y = \tan(1/2x - 1)$.

a. Period 2π, phase shift 0

b. Period 2π, phase shift 2 units left

c. Period π, phase shift 2 units right

d. **Period 2π, phase shift 2 units right ✓**

For $\tan(Bx - C)$: period $= \frac{\pi}{B} = 2\pi$, phase shift $= \frac{C}{B} = 2$ units right.

36 [Intermediate] • Target: Applications

A population of 500 grows at 5% per year, modeled by $P(t) = 500(1 + 0.05)^t$. Approximately how many years until the population reaches 1.5× its initial size (i.e. $P = 1.5 \cdot 500$)?

a. **8.3 years ✓**

b. 10.8 years

c. 6.3 years

d. 10.3 years

Solve $1.5 = (1 + 0.05)^t$: $t = \frac{\ln 1.5}{\ln(1 + 0.05)} \approx 8.3$ years.

37 [Intermediate] • Target: Comparing rates and estimating from a table

Given $f(x) = (x - 2)^2(x + 1)$, does the graph touch or cross the x-axis at $x = 2$?

a. The graph does not intersect the x-axis there

b. Not enough information

c. **Touches (does not cross) ✓**

d. Crosses

Multiplicity of $x = 2$ is 2 (even), so the graph touches the x-axis.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Average rate of change and secant lines • Cross-Lesson: 7.02 Logarithms and Their Properties

Find the reference angle for 220°.

a. 60°

b. **40°** ✓

c. 50°

d. 140°

220° lies in Quadrant 3; the reference angle is 40°.

39 [Challenging] • Target: Comparing average rate of change across different functions • Cross-Lesson: 7.03 Exponential and Logarithmic Equations

What are the maximum and minimum values of $y = 3\sin(x)$?

a. Max 2, Min -2

b. **Max 3, Min -3** ✓

c. Max 0, Min 0

d. Max 4, Min -4

Since $-1 \le \sin(x) \le 1$, y ranges from $-3 = -3$ to $3 = 3$.

40 [Challenging] • Target: Average rate of change from a graph • Cross-Lesson: 7.04 Applications: Growth and Decay

What is the amplitude and period of $y = 5\sin(3x) - 3$?

a. Amplitude -3, period $\frac{2}{3}\pi$

b. **Amplitude 5, period $\frac{2}{3}\pi$** ✓

c. Amplitude 5, period $\frac{3}{2}\pi$

d. Amplitude 5, period 2π

Amplitude $= |A| = 5$. Period $= \frac{2\pi}{3} = \frac{2}{3}\pi$.

41 [Challenging] • Target: Units and interpretation in context • Cross-Lesson: 8.01 Rational Functions and Asymptotes

Write $5^4 = 625$ in logarithmic form.

a. $\log_{625}(5) = 4$

b. $\log_5(4) = 625$

c. $\log_4(5) = 625$

d. $\log_5(625) = 4$ ✓

$b^x = y \Leftrightarrow \log_b(y) = x$, so $\log_5(625) = 4$.

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42 [Challenging] • Target: Computing average rate of change • Cross-Lesson: 8.02 Graphing Rational Functions

Convert 150° to radians.

a. $\frac{5\pi}{6}$ ✓

b. $\frac{2\pi}{3}$

c. $\frac{\pi}{6}$

d. $\frac{7\pi}{4}$

$150^\circ \times \frac{\pi}{180^\circ} = \frac{5\pi}{6}$.

43 [Challenging] • Target: Applications • Cross-Lesson: 8.03 Introduction to Limits

A principal of 1000 earns compound interest at rate 10% per period for 2 period(s), compounded annually. Find the final amount.

a. 1215

b. 1200

c. **1210** ✓

d. 1000

$A = 1000(1 + \frac{1}{10})^2 = 1210$.

44 [Challenging] • Target: Comparing rates and estimating from a table • Cross-Lesson: 9.01 Arithmetic Sequences and Series

Consider the unit circle shown, where a point $P(x, y)$ lies on the circle in the first quadrant. A ray from the origin through P forms an angle θ with the positive x-axis. If the y-coordinate of P is $\frac{\sqrt{3}}{2}$ and $\cos\theta = \frac{1}{2}$, find the value of x.

a. $\frac{\sqrt{3}}{2}$

b. $\frac{1}{2}$ ✓

c. 1

d. $\frac{\sqrt{2}}{2}$

The equation of a unit circle centered at the origin is $x^2 + y^2 = 1$. We are given that the y-coordinate of point P is $y = \frac{\sqrt{3}}{2}$. Substituting this into the equation of the unit circle: $x^2 + (\frac{\sqrt{3}}{2})^2 = 1$ $x^2 + \frac{3}{4} = 1$ $x^2 = 1 - \frac{3}{4}$ $x^2 = \frac{1}{4}$ Since point P lies in the first quadrant, its x-coordinate must be positive. Therefore, $x = \sqrt{\frac{1}{4}} = \frac{1}{2}$. Alternatively, we are given that the angle θ formed by the ray from the or.

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45 [Challenging] • Target: Average rate of change and secant lines • Cross-Lesson: 9.02 Geometric Sequences and Series

A circle is centered at the origin and passes through the point $P(a, 4)$ in the first quadrant. The equation of a tangent line to the circle at point P is given by $3x + 4y = k$. Find the value of k.

a. 25 ✓

b. 30

c. 35

d. 40

The equation of a circle centered at the origin is $x^2 + y^2 = r^2$. Since point $P(a, 4)$ lies on the circle, we have $a^2 + 4^2 = r^2$, so $a^2 + 16 = r^2$. The radius from the origin to $P(a, 4)$ is perpendicular to the tangent line at P. The slope of the radius is $m_{radius} = \frac{4-0}{a-0} = \frac{4}{a}$. The given tangent line is $3x + 4y = k$, which can be rewritten in slope-intercept form as $y = -\frac{3}{4}x + \frac{k}{4}$. The slope of the tangent line is $m_{tangent} = -\frac{3}{4}$. Since the rad.

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